

CHAPTER 16

EXERCISE TECHNIQUE FOR FREE WEIGHT AND MACHINE TRAINING

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After completing this chapter, you will be able to

- understand the general techniques involved in properly performing resistance training exercises;
- provide breathing guidelines;
- determine the appropriateness of wearing a weight belt;
- provide recommendations for spotting free weight exercises;
- teach proper resistance training exercise and spotting techniques; and
- identify common errors and flaws in technique.

This chapter provides guidelines and strategies for performing and teaching safe and effective lifting and spotting techniques. At the core of safe and effective resistance training is proper exercise

execution. Exercises that are performed and spotted correctly promote injury-free results and do so in a time-efficient manner.

The first half of this chapter summarizes the fundamental techniques involved in properly performing and spotting exercises and using a weight belt during lifting. The last section provides checklists and photographs depicting proper resistance training exercise and spotting techniques. It is assumed that the reader is familiar with these exercises. Therefore, the techniques presented are simply guidelines that represent the most commonly accepted method of performing the exercises. No attempt has been made to describe all variations for properly executing and spotting the exercises included in this chapter.

Fundamentals of Exercise Technique

Several commonalities are observed among resistance training exercise techniques. Most free weight and machine exercises involve some sort of handgrip on a bar, dumbbell, or handle, and absolutely all exercises require an optimal body or limb position, movement range and speed, and method of breathing. Additionally, some exercises may warrant the use of a weight belt and certain procedures for lifting a bar off the floor.

Handgrips

Two common grips are used in resistance training exercises: (a) the **pronated grip**, with palms down and knuckles up, also called the **overhand grip**; and (b) the **supinated grip**, with palms up and knuckles down, also known as the **underhand grip** ([figure 16.1](#)). (A variation of either grip is the **neutral grip**, in which the knuckles point laterally—as in a handshake.) Two less common grips are the **alternated grip**, in which one hand is in a pronated grip and the other is in a supinated grip, and the **hook grip**, which is similar to the pronated grip except that the thumb is positioned under the index and middle fingers. The hook grip is typically used for performing exercises that require a stronger grip (power exercises, e.g., snatch). Note that the thumb is wrapped around the bar in all of the grips shown; this positioning is called a **closed grip**. When the thumb does not wrap around the bar, the grip is called an open or **false grip**.

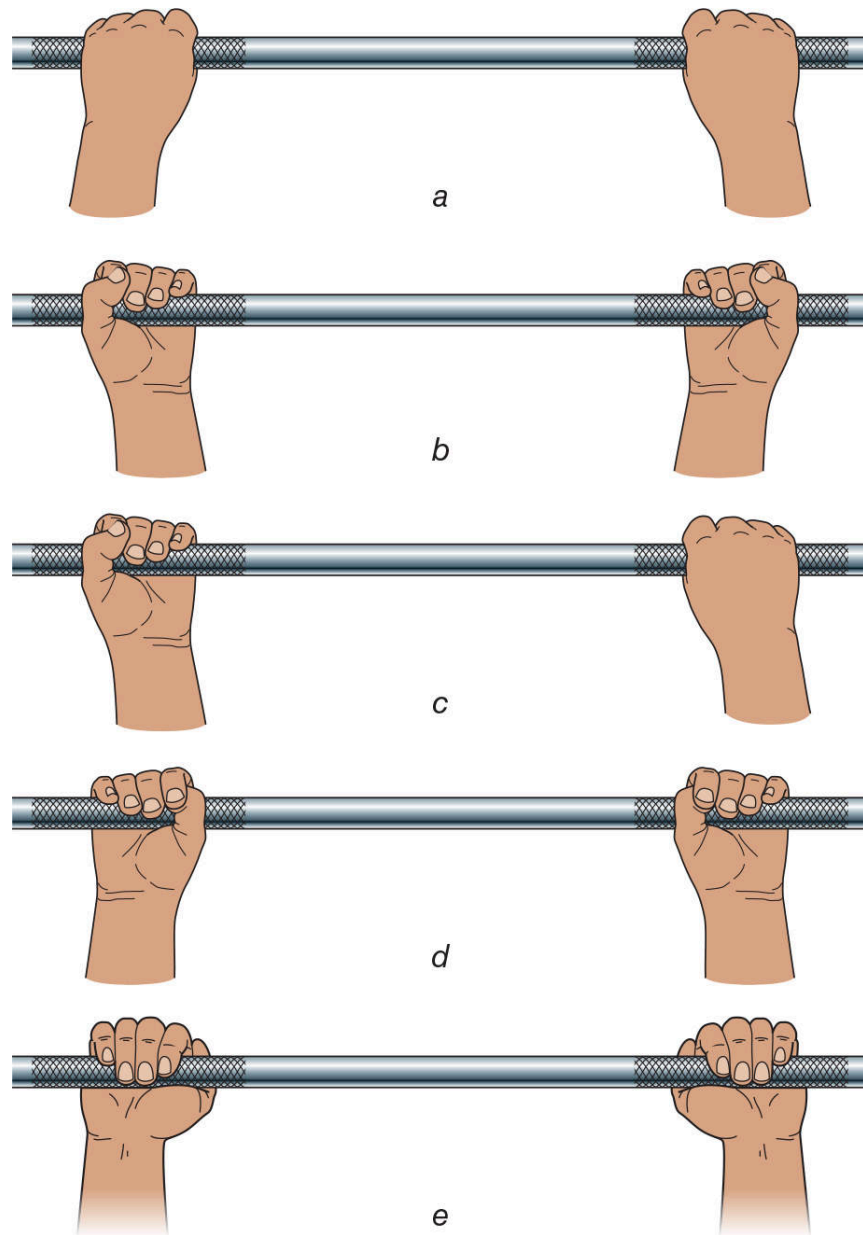


FIGURE 16.1 Bar grips: (a) pronated, (b) supinated, (c) alternated, (d) hook (posterior view), and (e) false.

Establishing the proper grip in an exercise involves placing the hands at the correct distance from each other (referred to as the **grip width**) and from the center of the bar (i.e., to maintain balance). The three grip widths ([figure 16.2](#)) are common, wide, and narrow. For most exercises, the hands are placed approximately shoulder-width apart. The hand positioning for all exercises should

result in a balanced, even bar. Weightlifting exercises and their variations use the **clean grip** or the **snatch grip**, depending upon the exercise being performed. Both of these grips are pronated closed hand positions. The clean grip is slightly wider than shoulder-width apart, outside of the knees. The snatch grip is a wide grip and can be determined using two types of measurements: the fist-to-opposite-shoulder method and the elbow-to-elbow method (also known as the *scarecrow method*); see [page 469](#). Both the clean grip and the snatch grip are often used with a hook grip (mentioned previously) to assist in giving the athlete a stronger grip.

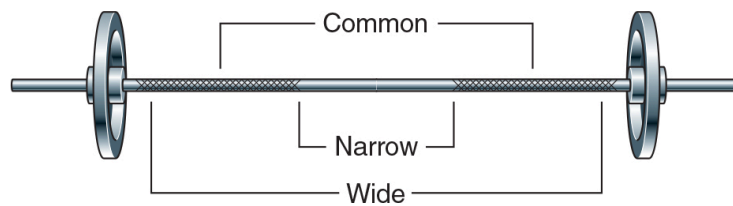


FIGURE 16.2 Grip widths.

Stable Body and Limb Positioning

Whether an exercise requires lifting a barbell or dumbbell from the floor or pushing and pulling while one is positioned in or on a machine, establishing a stable position is critical for safety and optimal performance. A stable position enables the athlete to maintain proper body alignment during an exercise, which in turn places an appropriate stress on muscles and joints.

Exercises performed while standing typically require that the feet be positioned slightly wider than hip width with the heels and balls of the feet in contact with the floor. Establishing a stable position in or on machines sometimes requires adjusting the seat or resistance arm and fastening belts snugly.

Seated or **supine** (lying face up) exercises performed on a bench require a specific posture. The athlete should position the body to achieve a **five-point body contact position**:

1. Head is placed firmly on the bench or back pad.

2. Shoulders and upper back are placed firmly and evenly on the bench or back pad.
3. Buttocks are placed evenly on the bench or seat.
4. Right foot is flat on the floor.
5. Left foot is flat on the floor.

Establishing and maintaining this five-point body contact position at the beginning and throughout the movement phases promotes maximal stability and spinal support (2, 9, 10).

Exercises performed during standing typically require that the feet be positioned slightly wider than hip width with the heels and balls of the feet in contact with the floor. Seated or supine exercises performed on a bench usually require a five-point body contact position.

Cam-, pulley-, or lever-based exercise machines that have an axis of rotation require specific positioning of the athlete's body, arms, or legs for reasons of safety and optimal execution. To align the primary joint of the body involved in the exercise with the axis of the machine, it may be necessary to move the seat; the ankle or arm roller pad; or the thigh, chest, or back pad. For example, adjust the ankle roller pad (up or down) and the back pad (forward or backward) to line up the knee joint with the machine axis before performing the leg (knee) extension exercise.

Range of Motion and Speed

When the entire **range of motion (ROM)** is covered during an exercise, the value of the exercise is maximized and flexibility is maintained or improved. Ideally, an exercise's full ROM should mimic the full ROM of the involved joint or joints in order for the greatest improvements to occur, but sometimes this is not possible (the knee of the trailing leg during a lunge) or recommended (using intervertebral joints during a squat).

Repetitions performed in a slow, controlled manner increase the likelihood that full ROM can be reached. However, when power or quick-lift exercises (e.g., power clean, push jerk, and snatch) are performed, an effort should be made to accelerate the bar to a maximal speed while still maintaining control and proper form throughout the exercise.

Breathing Considerations

The most strenuous movement of a repetition—typically soon after the transition from the eccentric phase to the concentric phase—is referred to as the **sticking point**. Strength and conditioning professionals should typically instruct athletes to exhale *through* the sticking point and to inhale during the less stressful phase of the repetition (3, 6, 7). For example, since the sticking point of the biceps curl exercise occurs about midway through the upward movement phase (concentric elbow flexion), the athlete should exhale during this portion. Inhalation, then, should occur as the bar is lowered back to the starting position. This breathing strategy applies to most resistance training exercises.

Breath holding may be suggested in some situations, however. For experienced resistance-trained athletes performing **structural exercises** (those that load the vertebral column and therefore place stress on it) with high loads, the **Valsalva maneuver** can be helpful for maintaining proper vertebral alignment and support. As explained in [chapter 2](#), the Valsalva maneuver involves expiring against a closed glottis, which, when combined with contracting the abdomen and rib cage muscles, creates rigid compartments of fluid in the lower torso and air in the upper torso (i.e., the “fluid ball”). The advantage of the Valsalva maneuver is that it increases the rigidity of the entire torso to aid in supporting the vertebral column, which in turn reduces the associated compressive forces on the disks during lifting (1, 4, 5, 8). It also helps to establish and maintain a normal lordotic lumbar spine position (also called a **neutral spine**) and erect upper torso position, described in the technique checklists

for certain exercises. Be aware, however, that the resulting increase in intra-abdominal pressure has potentially detrimental side effects, such as dizziness, disorientation, excessively high blood pressure, and blackouts. This is why the breath-holding phase is—and should be—quite transient, only about 1 to 2 seconds (at most). Even a very well-trained individual should not extend the length of the breath-holding phase, because blood pressure can quickly rise to triple resting levels (7).

Strength and conditioning professionals involved in conducting 1-repetition maximum (1RM) tests in, for example, the squat, deadlift, hip sled, leg press, shoulder press, or power clean need to be aware of the advantages and disadvantages of coaching athletes in the Valsalva maneuver. While it is obviously important that the vertebral column be internally supported during these movements for safety and technique reasons, it is recommended that an athlete not extend the breath-holding period.

For most exercises, exhale through the sticking point of the concentric phase and inhale during the eccentric phase. Experienced and well-trained athletes may want to use the Valsalva maneuver when performing structural exercises to assist in maintaining proper vertebral alignment and support.

Weight Belts

The use of a weight belt may help maintain intra-abdominal pressure during lifting (5, 7). Its appropriateness depends on the type of exercise performed and the relative load lifted. It is recommended that a weight belt be worn for exercises that place stress on the lower back and during sets that use near-maximal or maximal loads. Adopting this strategy may reduce the risk of lower back injury when combined with proper lifting and spotting techniques. A drawback to weight belt use is that wearing a belt too often reduces opportunities for the abdominal and lower back muscles to be

trained. Furthermore, no weight belt is needed for exercises that do not stress the lower back (e.g., biceps curl or lat pulldown) or for exercises that do stress the lower back (e.g., back squat or deadlift) but involve the use of light loads.

Typically an athlete should wear a weight belt when performing exercises that place stress on the lower back and during sets that involve near-maximal or maximal loads. A weight belt is not needed for exercises that do not stress the lower back or for those that do stress the lower back but involve light loads.

Spotting Free Weight Exercises

A **spotter** is someone who assists in the execution of an exercise to help protect the athlete from injury. A spotter may also serve to motivate the athlete and help in the completion of **forced repetitions** (also referred to as **partner-assisted reps**), but the spotter's primary responsibility is the safety of the athlete being spotted. The spotter must realize that poor execution of this responsibility may result in serious injury, not only to the athlete being spotted but also to the spotter and other individuals in close proximity. Although partner-assisted actions are valuable in helping an athlete benefit from training, the importance of promoting safety cannot be overemphasized.

The remainder of this section offers additional insight on when and how to spot free weight exercises. This information provides a foundation for strength and conditioning professionals to apply to their specific training environment.

Types of Exercises Performed and Equipment Involved

Free weight exercises performed over the head (e.g., barbell shoulder press) or with the bar on the back (e.g., back squat), racked anteriorly on the shoulders or on the clavicles (e.g., front squat) or over the face (e.g., bench press or lying triceps extension), are more challenging for the athlete to correctly execute than those in which the bar or dumbbells are held or raised at the sides or in front (e.g., lateral shoulder raise or barbell biceps curl, respectively) and therefore should involve one or more spotters. The overhead, bar-on-the-back or bar-on-the-front-shoulders and over-the-face exercises (especially using dumbbells) also require more skill on the part of the spotter and are potentially the most dangerous to the athlete. Spotting dumbbell exercises typically requires more skill than spotting barbell exercises because there is an additional piece of

equipment to observe and spot. **Power exercises** should not be spotted.

With the exception of power exercises, free weight exercises performed with a bar moving over the head, positioned on the back, racked on the front of the shoulders, or passing over the face typically require one or more spotters.

Spotting Overhead Exercises and Those With the Bar on the Back or Front Shoulders

Ideally, to promote the safety of the athlete, the spotters, and others nearby, overhead exercises and those involving the bar on the back or front shoulders should be performed inside a power rack with the crossbars in place at an appropriate height. All plates, bars, locks, and weight-plate trees must be cleared from the lifting area so they cannot be tripped over or run into and cannot affect the movement of the bar. Athletes who are not lifting should be instructed to stay clear of the lifting area. Since the loads lifted in these exercises can be considerable, to exert sufficient leverage the spotter or spotters should be at least as strong and at least as tall as the athlete who is lifting. **Out-of-the-rack exercises** (e.g., forward step lunge or step-up) with heavy weights can result in serious injury. These exercises should be executed only by well-trained and skilled athletes and spotted by experienced professionals.

Spotting Over-the-Face Exercises

When spotting **over-the-face barbell exercises**, it is important for the spotter to grasp the bar with an alternated grip, usually narrower than the athlete's grip. Since the spotter may be called on to catch the bar or assist in lifting moderate to heavy loads (usually from a higher vantage point), establishing a solid, wide base of support with the feet and a neutral spine position is critically important.

For dumbbell exercises, it is important to spot as close to the dumbbells as possible or, in a few exercises, to spot the dumbbell itself. Although some individuals advocate spotting dumbbell movements by placing the hands on the athlete's upper arms or elbows ([figure 16.3a](#)), this technique may lead to injury. If the athlete's elbows "collapse" (i.e., flex), the spotter will not be in a position to stop the dumbbells from striking the athlete's face or chest. Spotting at the forearms near the wrists ([figure 16.3b](#)) is a safer technique. Note that for some exercises (e.g., dumbbell pullover and overhead dumbbell triceps extension), it is necessary to spot with hands on the dumbbell itself.

Do Not Spot Power Exercises

Whereas spotting is recommended in the types of exercises previously described, it is not advised in power exercises. Instead of spotting these exercises, the strength and conditioning professional needs to teach athletes how to get away from a bar that is unmanageable. Spotting these types of exercises is too dangerous to both the spotter and athlete. Athletes should be instructed that when they miss the bar in front, they should push the bar away or simply drop it. They should be taught that if they lose the bar behind the head, they should release it and jump forward. For these reasons, the surrounding area or platform should be cleared of other athletes and equipment before such exercises are performed.



FIGURE 16.3 (a) Incorrect dumbbell spotting location. (b) Correct spotting location.

Number of Spotters

The number of spotters needed is largely determined by the load being lifted, the experience and ability of the athlete and spotters, and the physical strength of the spotters. With heavier loads, the likelihood and severity of an injury increase. Once the load exceeds the spotter's ability to effectively protect the athlete (and him- or herself), another spotter must become involved. On the other hand, one spotter is preferred if he or she can easily handle the load, because two or more spotters must coordinate their actions with those of the athlete. As the number of spotters increases, so does the chance that an error in timing or technique may occur.

Communication Between Athlete and Spotter

Communication is the responsibility of both the spotter and the athlete. Before beginning a set, the athlete should tell the spotter how the bar will initially be handled, how many repetitions will be performed, and when he or she is ready to move the bar into position. If spotters do not have this information, they may take

control of the bar improperly, too soon, or too late and consequently disrupt the exercise or injure the athlete.

Use of a Liftoff

The term **liftoff** refers to moving the bar from the upright supports to a position in which the athlete can begin the exercise. Usually the spotter helps place the barbell or dumbbells into the athlete's hands while the elbows are extended and helps to move the barbell or dumbbells to the proper starting position. Some athletes may want the spotter to provide a liftoff; others do not. If a liftoff is needed or requested, the athlete and spotter need to agree in advance on a verbal signal (e.g., the command "up" or "I will give it to you on the count of three"). Typically, the athlete signals that he or she is ready; the spotter says, "One, two, three," and on "three" the bar is moved into position. Liftoffs are normally used in the bench press (off the supports), shoulder press (off the supports or the shoulders), and squatting exercises if the supports are too low and nonadjustable. When two spotters are involved and the athlete wants a liftoff, as with a bench press, one spotter should assist with the liftoff and then quickly move to the end of the bar to spot (the other spotter is already at the other end). The spotter providing the liftoff must be sure that the athlete has complete control of the bar at the conclusion of the liftoff. If two spotters are used, both should help the athlete place the bar back onto the supports on completion of the exercise.

Amount and Timing of Spotting Assistance

Knowing how much and when to help an athlete is an important aspect of spotting, and it requires the spotter to be experienced in proper spotting techniques. Most athletes typically need just enough help to successfully complete a repetition through a sticking point (i.e., a partner-assisted action); other times they might need the spotter to handle the entire load. At the first indication that a repetition will be missed, the athlete should quickly ask or signal the

spotter (sometimes with just a grunt or sound) for help, and the spotter needs to immediately attempt to provide the amount of assistance needed. If the athlete cannot contribute anything to the completion of the repetition, the athlete should immediately tell the spotter to “take it” or use some other phrase agreed on before the lift. Regardless of when or why the spotter is needed to assist, he or she should take the bar—if possible—from the athlete quickly and smoothly, trying to avoid abrupt changes in the amount of load being handled by the athlete. The athlete should try to stay with the bar until it is racked or placed safely on the floor. This helps to protect both the spotter and athlete from injury.

The spotting guidelines provided in the exercise technique checklists are appropriate for a typical training environment. Spotting procedures may vary when excessively heavy loads are being used—such as 1RM attempts—because more spotters are typically needed.

Guidelines for Spotting

Certain exercises that require spotting (over the face or neck and on the back) have a higher level of difficulty and thus warrant additional safety considerations. In any exercise where the potential for injury might occur, spotting should be used.

- Spotting is necessary on skillful multijoint exercises that involve a barbell or dumbbells being lifted over the head, on the back, on the front of the shoulders, or over the face.
- Exercises done overhead and with a bar racked on the back or in front of the shoulders may require more than one spotter, and multiple spotters must be taught how and where to spot the exercise.
- Spotting over-the-face exercises requires the spotter to grasp the implement being lifted or the athlete’s wrists. Spotters must be trained to grasp the implement properly and even assist athletes in reracking the bar or returning dumbbells to the start position in case an exercise becomes too difficult to complete safely on their own.
- Power exercises are not spotted due to the safety factors for the athlete and the spotter or spotters, and athletes should be taught how to miss properly when learning these exercises.
- Determining the number of spotters for certain exercises depends on a number of factors, especially the amount of load being used, because the likelihood of injury can increase greatly with heavier loads.

- Both spotters and athletes must communicate properly before beginning an exercise requiring increased safety. They both must take part in this communication and be clear as to what exercise is being performed and how many repetitions. Therefore, communicating on exercises that require a liftoff is critical to safety in these exercises.
- The liftoff requires a spotter to assist the athlete in getting the implement to the proper position before beginning the exercise and can require multiple spotters, depending on the exercise.

The spotting guidelines provided in this chapter should be a fundamental part of any strength and conditioning professional's training. Safety of athletes in the weight room is the most important part of supervising and conducting proper training programs and providing the best and safest atmosphere for athletes to be successful in training.

Conclusion

It is critically important that the strength and conditioning professional provide athletes with proper instruction in resistance training exercise technique, including the fundamentals of breathing and weight belt use. Additionally, the strength and conditioning professional needs to teach athletes how and when exercises should be spotted. Combined with quality supervision and ongoing feedback, this attention to proper instruction results in a safe training environment and an effective and appropriate training stimulus.

RESISTANCE TRAINING EXERCISES

Videos for exercises identified by this icon  can be found in *HKPropel*.

ABDOMEN

- 1 Bent-Knee Sit-Up [418](#)
- 2 Abdominal Crunch [418](#)
- 3 Abdominal Crunch (Machine) [419](#)

BACK

-  4 Bent-Over Row [420](#)
- 5 Pendlay Row [421](#)
- 6 One-Arm Dumbbell Row [422](#)
-  7 Lat Pulldown (Machine) [423](#)
-  8 Seated Row (Machine) [424](#)
-  9 Low-Pulley Seated Row (Machine) [425](#)

BICEPS

-  10 Barbell Biceps Curl [426](#)
- 11 Hammer Curl [427](#)

CALVES

12 Standing Calf (Heel) Raise (Machine) [428](#)

13 Seated Calf (Heel) Raise (Machine) [429](#)

CHEST

▶ **14** Flat Barbell Bench Press (and Dumbbell Variation) [430](#)

15 Incline Dumbbell Bench Press (and Barbell Variation) [432](#)

▶ **16** Flat Dumbbell Fly (and Incline Variation) [433](#)

▶ **17** Vertical Chest Press (Machine) [434](#)

18 Pec Deck (Machine) [435](#)

FOREARMS

19 Wrist Curl [436](#)

20 Wrist Extension [437](#)

HIPS AND THIGHS

▶ **21** Hip Sled (Machine) [438](#)

▶ **22** Back Squat [439](#)

▶ **23** Front Squat [441](#)

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▶ **25** Forward Step Lunge [445](#)

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28 Good Morning [451](#)

29 Deadlift [452](#)

30 Stiff-Leg Deadlift [453](#)

31 Romanian Deadlift (RDL) (and Snatch Grip Variation) [454](#)

▶ **32** Leg (Knee) Extension (Machine) [455](#)

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SHOULDERS

- ▶ **34** Shoulder Press (Machine) [457](#)
- ▶ **35** Seated Barbell Shoulder Press (and Dumbbell Variation) [458](#)
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TRICEPS

- ▶ **38** Lying Barbell Triceps Extension [461](#)
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POWER EXERCISES

- ▶ **40** Push Press [463](#)
- ▶ **41** Push Jerk [465](#)
- ▶ **42** Power Clean (and Hang Power Clean Variation) [467](#)
- ▶ **43** Power Snatch (and Hang Power Snatch Variation) [469](#)
- ▶ **44** Clean Grip Mid-Thigh Pull (and Snatch Grip Variation) [471](#)
- ▶ **45** Clean Pull from Knee-Block (and Snatch Grip Variation) [473](#)

Abdomen

1 BENT-KNEE SIT-UP

Starting Position

- Lie in a supine position on a floor mat.
- Flex the knees to bring the heels near the buttocks.
- Fold the arms across the chest or abdomen.
- All repetitions begin from this position.

Upward Movement Phase

- Flex the neck to move the chin toward the chest.
- Keeping the feet, buttocks, and lower back neutral and stationary on the mat, curl the torso toward the thighs until the upper back is off the mat.

Downward Movement Phase

- Uncurl the torso back to the starting position.
- Keep the feet, buttocks, lower back, and arms in the same position.

MAJOR MUSCLE INVOLVED

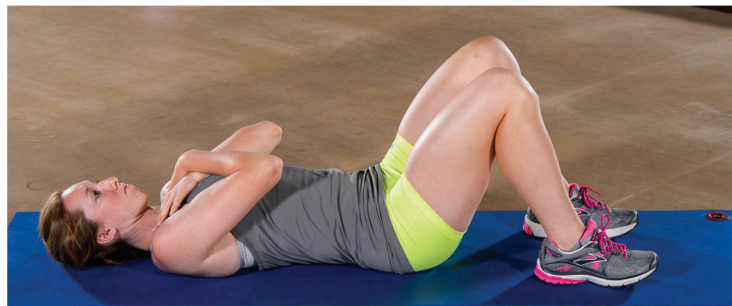
rectus abdominis

VERBAL CUES

- Curl torso toward thighs
- Maintain feet on the floor

COMMON ERRORS

- Not keeping the feet flat on the floor throughout the movement
- Not lowering the back fully to the ground



Starting position



Upward and downward movements

2

ABDOMINAL CRUNCH

Starting Position

- Lie in a supine position on a floor mat.
- Place the heels on a bench with the hips and knees flexed to about 90 degrees.
- Fold the arms across the chest or abdomen.
- All repetitions begin from this position.

Upward Movement Phase

- Flex the neck to move the chin toward the chest.
- Keeping the buttocks and lower back neutral and stationary on the mat, curl the torso toward the thighs until the upper back is off the mat.

Downward Movement Phase

- Uncurl the torso back to the starting position.
- Keep the feet, buttocks, lower back, and arms in the same position.

MAJOR MUSCLE INVOLVED

rectus abdominis

VERBAL CUES

- **Curl shoulders off floor**
- **Curl shoulders toward thighs**

COMMON ERRORS

- **Raising the heels off the bench during the upward movement phase**
- **Raising the hips off the floor during the downward movement phase**



Starting position



Upward and downward movements

3

ABDOMINAL CRUNCH (MACHINE)

Starting Position

- Sit down on the machine and press the back firmly against the back pad.
- Place the feet on the floor and the legs behind the roller pads.
- Position the legs parallel to each other.
- Reach back and grasp the handles with a closed, neutral grip with the back of the upper arms pressed against the arm pads.
- All repetitions begin from this position.

Forward Movement Phase

- Keeping the buttocks on the seat and the legs stationary, curl the torso forward toward the thighs.

Backward Movement Phase

- Uncurl the torso back to the starting position.

- Keep the feet, buttocks, lower back, and arms in the same position.

MAJOR MUSCLE INVOLVED

rectus abdominis

VERBAL CUES

- Curl shoulders toward thighs
- Maintain contact with the seat

COMMON ERRORS

- Raising the buttocks off the seat during the movement
- Overemphasizing a pull with the legs or hands to help curl the torso



Starting position



Forward and backward movements

Back



4

BENT-OVER ROW

Before Beginning

- Grasp the bar with a closed, pronated grip.
- Grip should be wider than shoulder-width.
- Lift the bar from the floor as described for the deadlift exercise. Use a pronated grip, not an alternated grip.

Starting Position

- Position the feet in a shoulder-width stance with the knees slightly flexed.
- Flex forward at the hips so the torso is slightly above parallel to the floor.
- Create a neutral spine position.
- Focus the eyes a short distance ahead of the feet.
- Allow the bar to hang with the elbows fully extended.
- All repetitions begin from this position.

Upward Movement Phase

- Pull the bar toward the torso.
- Keep the torso rigid, back neutral, and knees slightly flexed.
- Do not jerk the torso upward.
- Touch the bar to the lower chest or upper abdomen.

Downward Movement Phase

- Lower the bar back to the starting position.
- Maintain the neutral spine and stationary torso and knee positions.
- At the end of the set, flex the hips and knees to place the bar on the floor and stand up.

MAJOR MUSCLES INVOLVED

latissimus dorsi, teres major, middle trapezius, rhomboids, posterior deltoids

VERBAL CUES

- **Maintain a flat back (neutral spine)**

- Focus on a point ahead of your toes
- Squeeze a pencil between your shoulder blades

COMMON ERRORS

- Jerking the upper body, extending the torso, locking out the knees, curling bar in the hands, or rising on the toes to help move the bar toward the upper abdomen
- Allowing the upper back to round (losing the neutral spine position) during the movement



Starting position



Upward and downward movements

5

PENDLAY ROW

Starting Position

- Position the feet in a shoulder-width stance with the knees slightly flexed and the barbell in front.
- Flex at the hips and knees to lower the torso until it is parallel to the floor.
- Grasp the bar with a closed, pronated grip.
- Grip should be wider than shoulder width.

- Create a neutral spine position.
- Focus the eyes on the floor a short distance ahead of the feet.
- Allow the bar to rest on the floor with the elbows fully extended.
- All repetitions begin from this position.

Upward Movement Phase

- Pull the bar toward the torso.
- Keep the torso rigid, with back neutral and parallel to the floor and the knees slightly flexed.
- Do not jerk the torso upward.
- Touch the bar to the lower chest or upper abdomen.

Downward Movement Phase

- Lower the bar back to the starting position.
- Maintain the neutral spine, parallel torso, and flexed hip and knee positions.
- At the end of the set, place the bar on the floor and stand up.

MAJOR MUSCLES INVOLVED

latissimus dorsi, teres major, middle trapezius, rhomboids, posterior deltoids

VERBAL CUES

- **Elbows high and wide**
- **Reset on each rep**
- **Hips back, not up**

COMMON ERRORS

- **Allowing the upper back to round (losing the neutral spine position) during the movement**
- **Not maintaining a torso position that is parallel to the floor**



Starting position



Upward movement

6

ONE-ARM DUMBBELL ROW

Starting Position

- Position the feet in a shoulder-width stance with the knees slightly flexed.
- Flex forward at the hips so the torso is slightly above parallel to the floor.
- Create a neutral spine position.
- Grasp the dumbbell with a closed, neutral grip.
- Place the opposite hand on the bench for support.
- Allow the dumbbell to hang with the elbow fully extended.
- All repetitions begin from this position.

Upward Movement Phase

- Pull the dumbbell toward the torso, keeping the elbow close to the body.
- Keep the torso rigid, back neutral, and knees slightly flexed.
- Touch the dumbbell to the side of the torso.

Downward Movement Phase

- Lower the dumbbell back to the starting position.
- Maintain the neutral spine and stationary torso and flexed knee positions.

MAJOR MUSCLES INVOLVED

latissimus dorsi, teres major, middle trapezius, rhomboids, posterior deltoids

VERBAL CUES

- Lead with the elbow
- Pull toward the hip pocket
- Full stretch at the bottom of the movement

COMMON ERRORS

- Rounding the back and not maintaining a neutral spine
- Not fully extending the arm in the downward movement phase



Starting position



Upward and downward movements

7 LAT PULLDOWN (MACHINE)

Starting Position

- Grasp the bar with a closed, pronated grip.
- Grip should be wider than shoulder width.
- Sit down on the seat facing the machine.
- Position the thighs under the pads with the feet flat on the floor. If necessary, adjust the seat and thigh pad.
- Lean the torso slightly backward.
- Extend the elbows fully.
- All repetitions begin from this position.

Downward Movement Phase

- Pull the bar down and toward the upper chest.
- Maintain the slight torso backward lean; do not jerk the torso backward.
- Touch the bar to the clavicle and upper chest area.

Upward Movement Phase

- Allow the elbows to slowly extend back to the starting position.
- Keep the torso in the same position.
- At the end of the set, stand up and return the bar to its resting position.

MAJOR MUSCLES INVOLVED

latissimus dorsi, teres major, middle trapezius, rhomboids, posterior deltoids

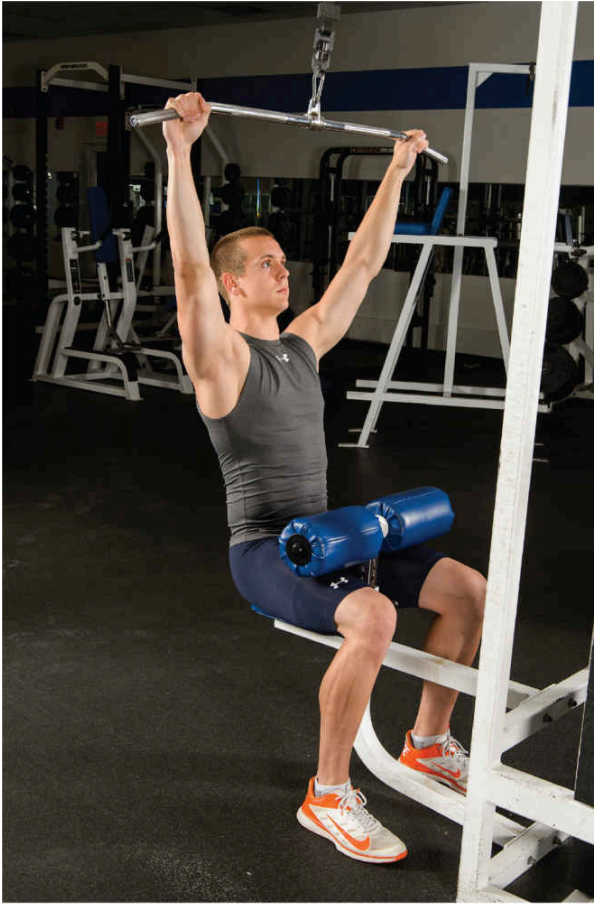
VERBAL CUES

- **Chest up, shoulders back**
- **Lead with your elbows**
- **Full stretch at the top**

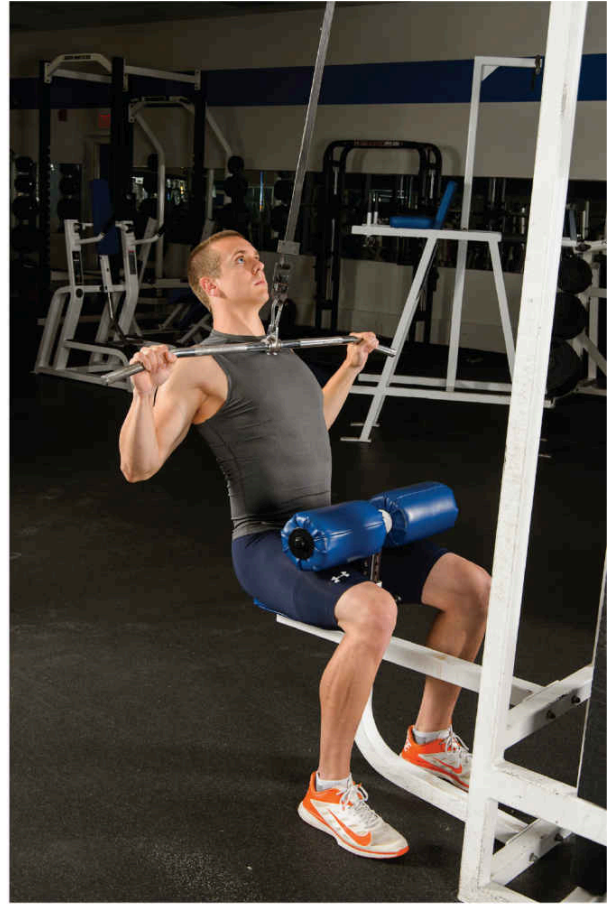
COMMON ERRORS

- **Excessive backward lean and failure to maintain a neutral spine**

- **Pulling the bar down behind the head to the back of the neck**



Starting position



Downward and upward movements

8 SEATED ROW (MACHINE)

Starting Position

- Sit erect with the feet flat on the platform and press the torso against the chest pad.
- Grasp the handles with a closed grip, either pronated or neutral. If necessary, adjust the seat height to position the arms approximately parallel to the floor.
- Allow the elbows to extend fully.
- All repetitions begin from this position.

Backward Movement Phase

- Pull the handles toward the chest or upper abdomen.
- Maintain an erect torso position and keep the elbows next to the torso.
- Pull the handles as far back as possible.
- Do not jerk the torso backward.

Forward Movement Phase

- Allow the handles to move forward, back to the starting position.
- Keep the torso in the same position.

MAJOR MUSCLES INVOLVED

latissimus dorsi, teres major, middle trapezius, rhomboids, posterior deltoids

VERBAL CUES

- **Sit tall**
- **Pull with your elbows**
- **Full stretch at the start**

COMMON ERRORS

- **Moving the chest off the pad**
- **Lifting the feet off the ground or not bracing against the footrests if available**



Starting position



Backward and forward movements



9

LOW-PULLEY SEATED ROW (MACHINE)

Starting Position

- Sit on the long seat pad (or on the floor if the seat pad is not available) and place the feet on the foot supports or the machine frame.
- Grasp the handles with a closed grip, either neutral or pronated.
- Sit in an erect position with the torso perpendicular to the floor, knees slightly flexed, and the feet and legs parallel to each other.
- Allow the elbows to fully extend with the arms approximately parallel to the floor.
- All repetitions begin from this position.

Backward Movement Phase

- Pull the handles toward the abdomen.
- Maintain an erect torso position with the knees in the same slightly flexed position. Do not jerk the upper body or lean back.

- Continue pulling until the handles touch the abdomen.

Forward Movement Phase

- Allow the elbows to slowly extend back to the starting position.
- Maintain an erect torso position with the knees in the same slightly flexed position.
- At the end of the set, flex the knees and hips to return the weight to its resting position.

MAJOR MUSCLES INVOLVED

latissimus dorsi, teres major, middle trapezius, rhomboids, posterior deltoids

VERBAL CUES

- **Sit tall and keep your chest up**
- **Full stretch at the start**
- **Pull handles to your belly button**

COMMON ERRORS

- **Jerking the upper body or leaning back during the backward movement phase**
- **Flexing the torso forward during the forward phase**



Starting position



Backward and forward movements

Biceps

10 BARBELL BICEPS CURL

Starting Position

- Grasp the bar with a closed, supinated grip.
- Grip should be shoulder width so the arms touch the sides of the torso.
- Stand erect with the feet shoulder-width apart and the knees slightly flexed.
- Rest the bar on the front of the thighs with the elbows fully extended.
- All repetitions begin from this position.

Upward Movement Phase

- Flex the elbows until the bar is near the anterior deltoids.
- Keep the torso erect and the upper arms stationary.
- Do not jerk the body or swing the bar upward.

Downward Movement Phase

- Lower the bar until the elbows are fully extended.

- Keep the torso and knees in the same position.
- Do not bounce the bar on the thighs between repetitions.

MAJOR MUSCLES INVOLVED

biceps brachii, brachialis, brachioradialis

VERBAL CUES

- **Pin your elbows to your sides**
- **Curl through the full range**
- **Control the descent**

COMMON ERRORS

- **Jerking the upper body, shrugging the shoulders, extending the torso, extending the knees, swinging the bar, or rising on the toes to help raise the bar**
- **Keeping the elbows partially flexed at the end of the downward movement phase (shortening the ROM)**